



FSE+ Fondo Social Europeo Plus

OlvidAI 

CRISTINA ÁLVAREZ IZQUIERDO
JULIA JIMÉNEZ ABELLEIRA
PABLO MARTINEZ ABRIL
ÁNGEL PORRÓN COSTO



Cofinanciado por
la Unión Europea



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DE INDUSTRIA
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EOI

Escuela de
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SAMSUNG



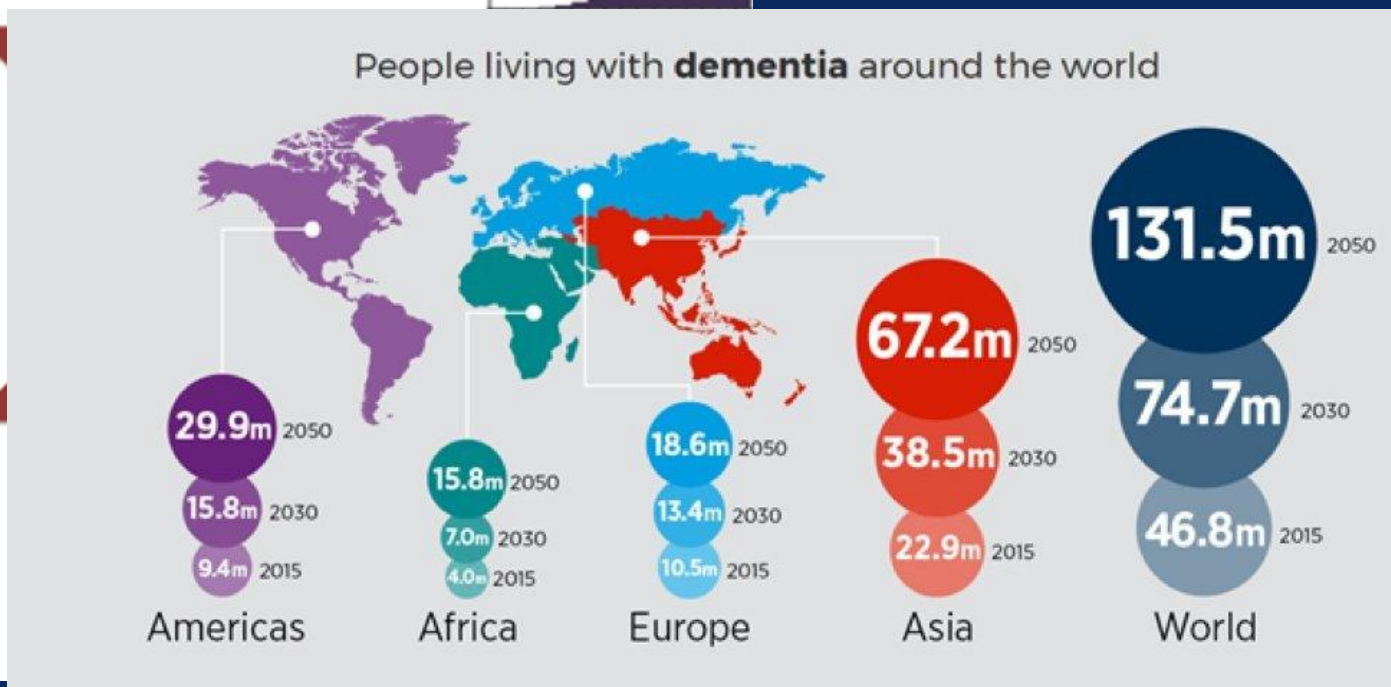
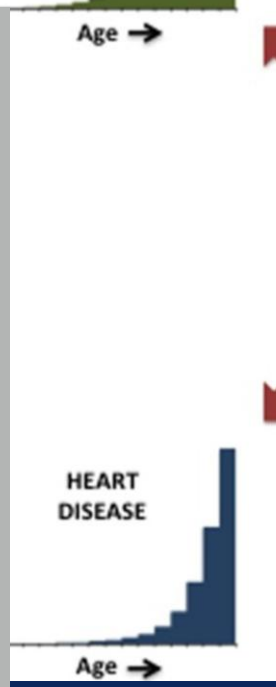
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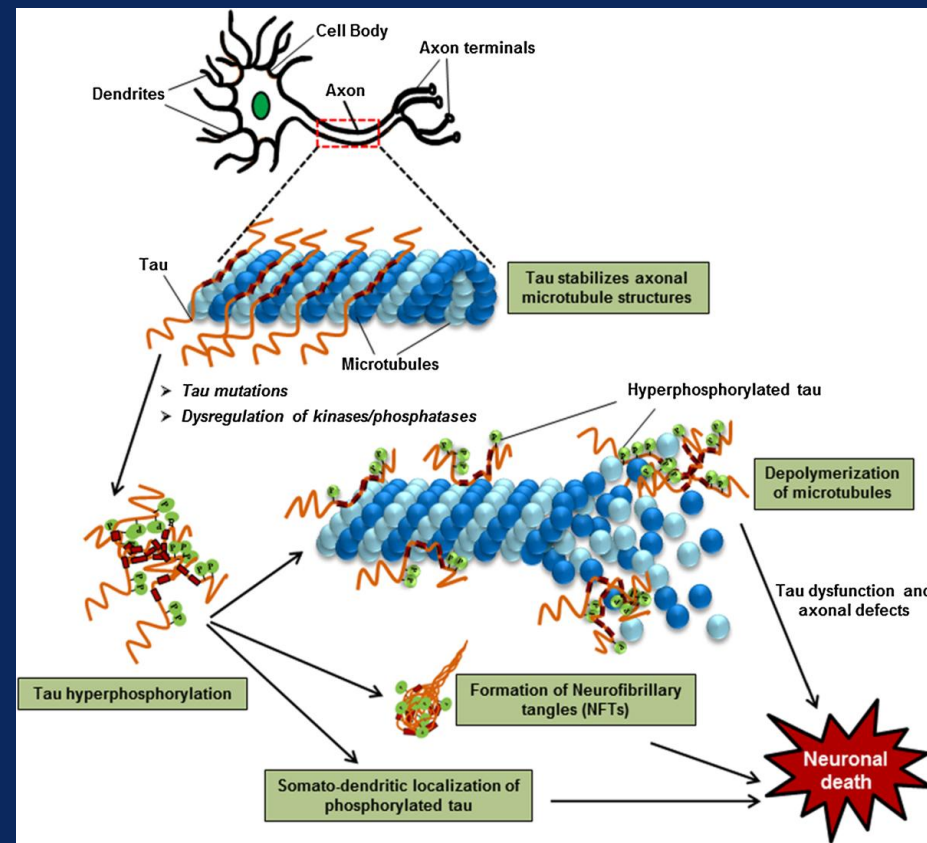
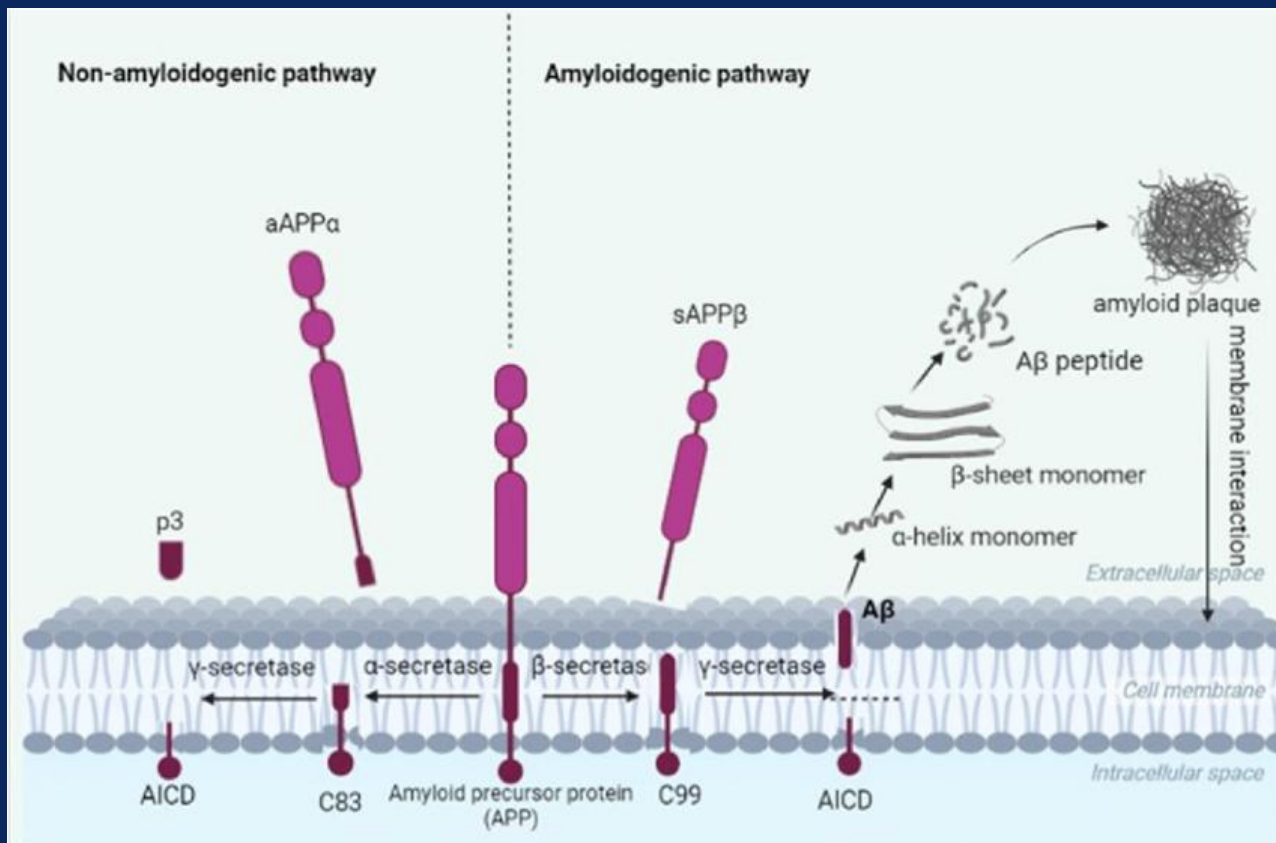


1. INTRODUCCION





1.1 Fisiopatología de la enfermedad de Alzheimer



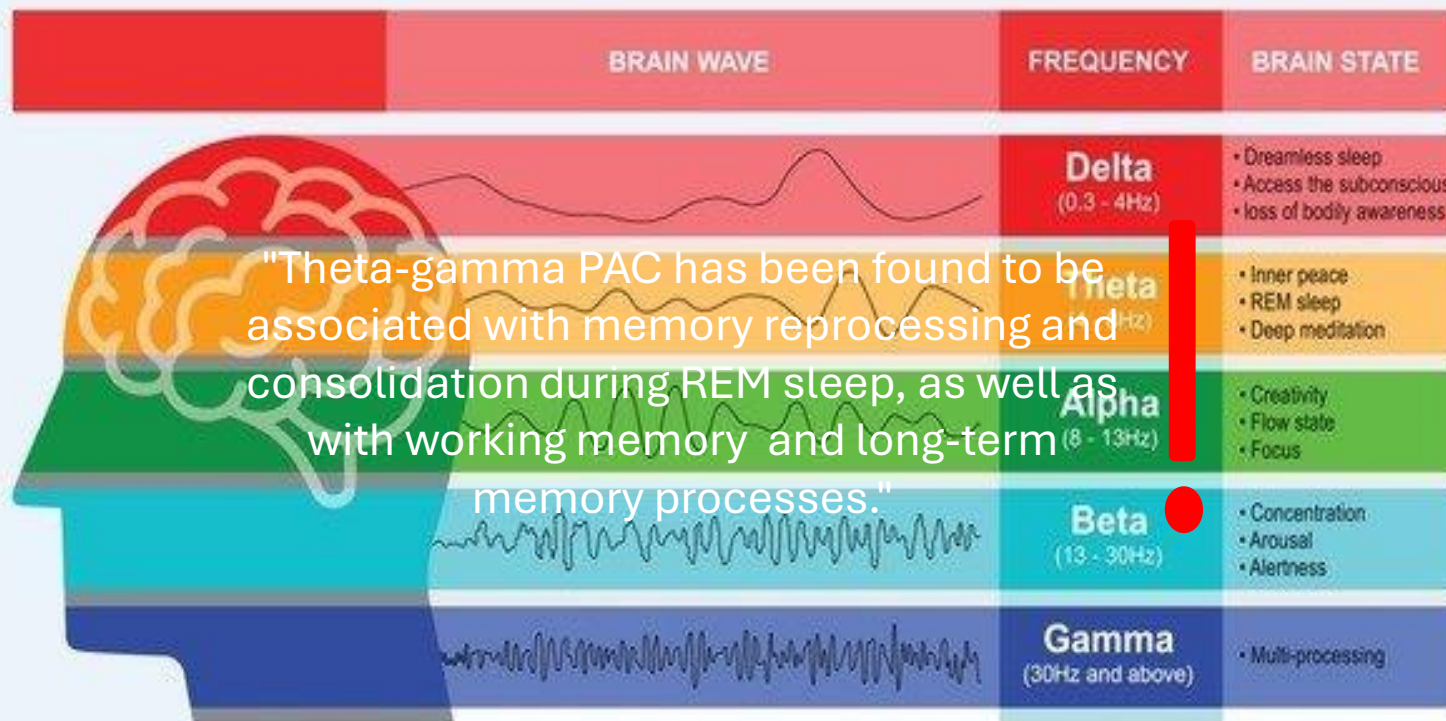
BIOMARCADORES PARA LA DETECCIÓN DEL ALZHEIMER



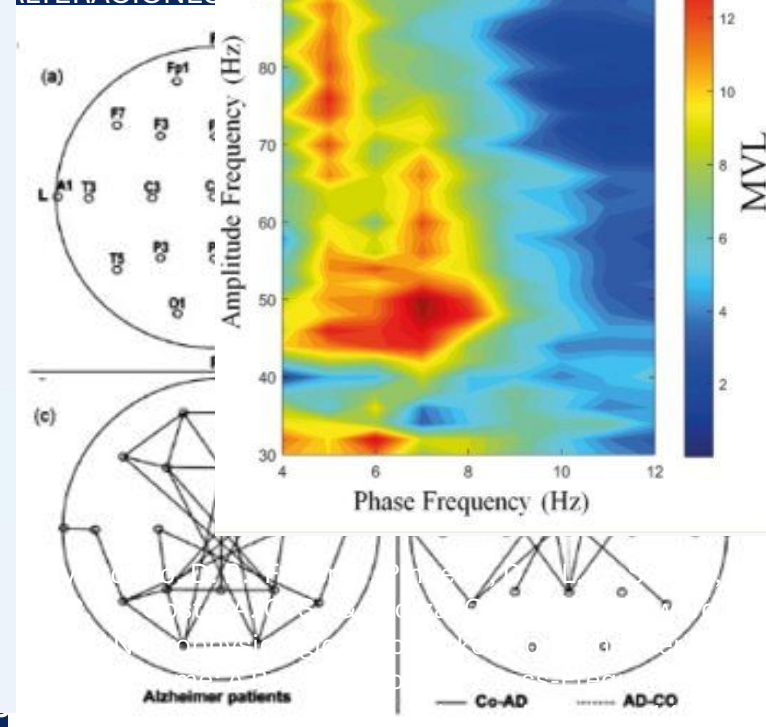


1.2 Redes neuronales y conectividad funcional

"Theta-gamma PAC has been found to be associated with memory reprocessing and consolidation during REM sleep, as well as with working memory and long-term memory processes."



Jones, G N. *Functional Connectivity in the Human Brain*. January 2007. https://doi.org/10.1007/978-1-4020-5488-8_12



Functional Networks
Volume 17,
r/bhj127

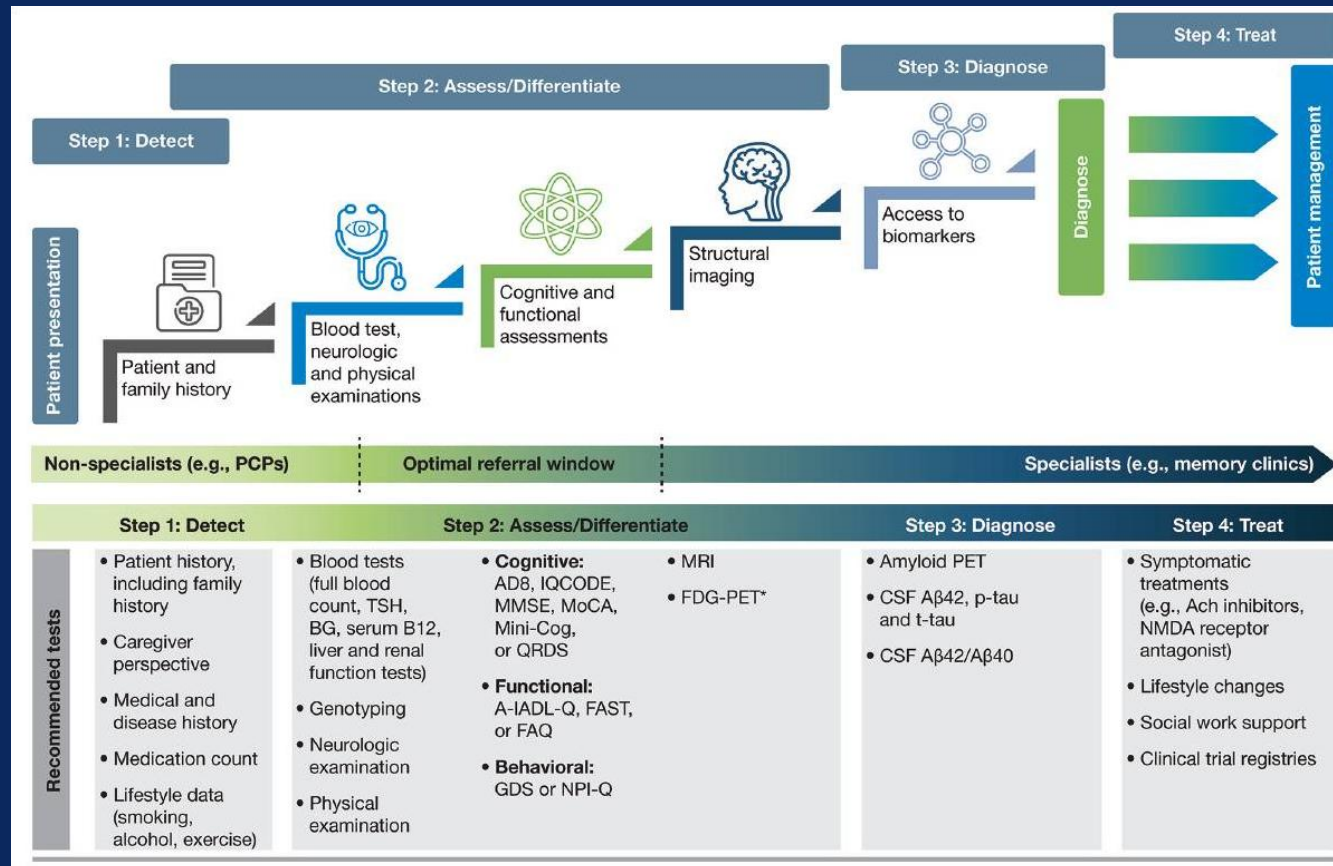
Almeida, A.
Identification
se in Down
Amplitude
449.

*Patrones anatómicos y de conectividad entre regiones distantes
Palop, J., Mucke, L. Amyloid- β -induced neuronal dysfunction in Alzheimer's disease: from synapses toward neural networks. *Nat Neurosci* **13**, 812–818 (2010). <https://doi.org/10.1038/nn.2583>

<https://doi.org/10.14336/AD.2022.0906>



1.3 Detección, diferenciación y diagnóstico



Porsteinsson, A. P., Isaacson, R. S., Knox, S., Sabbagh, M. N., & Rubino, I. (2021). Diagnosis of Early Alzheimer's Disease: Clinical Practice in 2021. *The journal of prevention of Alzheimer's disease*, 8(3), 371–386. <https://doi.org/10.14283/jpad.2021.23>



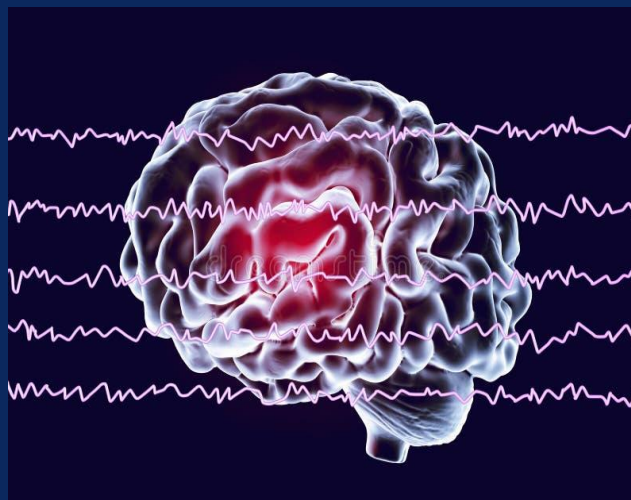
2. MODELO DE NEGOCIO



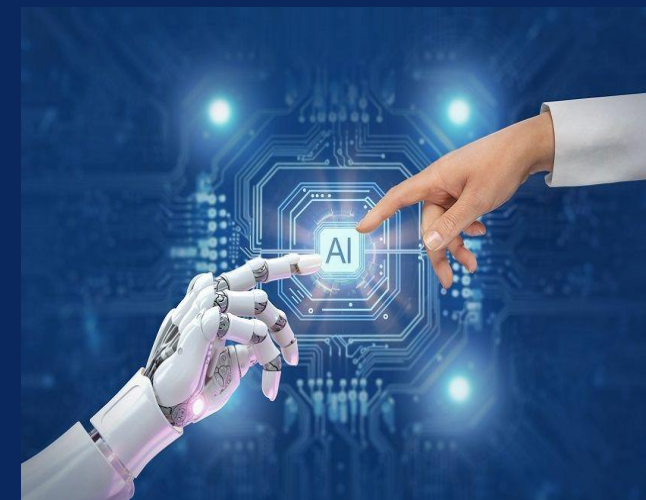
- *Modelo B2B en Salud Digital basado en EEG + IA*
- *Formato de venta: suscripción (SaaS) o licencia de software*



+



+

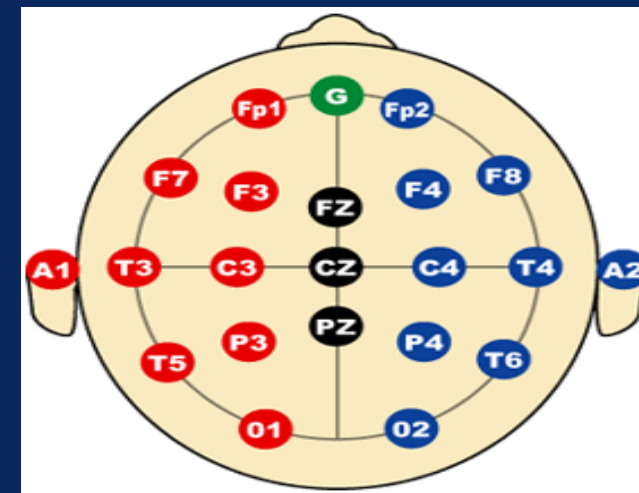




3. DATOS Y VARIABLES



- *Datos utilizados:*
 - *Registros de EEG de pacientes con y sin Alzheimer (Edad comprendida entre 60 y 70 años)*
 - *64 sujetos -> 35 Alzheimer vs 29 Control*
 - *19 canales estándar del EEG (Fp1, Fp2, F3, F4, F7, F8, Cz, Pz, O1, O2, etc.)*
 - *Etiquetas: Diagnosis (0 = Control, 1 = Alzheimer)*





```
# =====
```

```
# 1. Librerías
```

```
# =====
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.model_selection import train_test_split, GridSearchCV
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.metrics import (  
    accuracy_score, classification_report, confusion_matrix,  
    roc_auc_score, roc_curve
```

```
)
```

```
import warnings
```

```
warnings.filterwarnings("ignore")
```

```
# =====
```

```
# 2. Cargar datos
```

```
# =====
```

```
df = pd.read_csv("alzheimers_disease_data.csv")
```

```
df["AD"] = df["Diagnosis"] == 1
```

```
df["No-AD"] = df["Diagnosis"] == 0
```

```
cols = [
```

```
    'MMSE', 'FunctionalAssessment',
```

```
    'MemoryComplaints', 'BehavioralProblems', 'ADL',
```

```
    'FamilyHistoryAlzheimers',
```

```
    'Diagnosis', 'AD', 'No-AD'
```

```
]
```

```
df = df[cols]
```

```
X = df.drop(columns=["AD", "No-AD", "Diagnosis"])
```

```
y = df["AD"]
```

```
# Escalado opcional (Random Forest no lo requiere, pero lo dejamos para consistencia)
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X_scaled, y, test_size=0.3, random_state=42, stratify=y
```

```
)
```



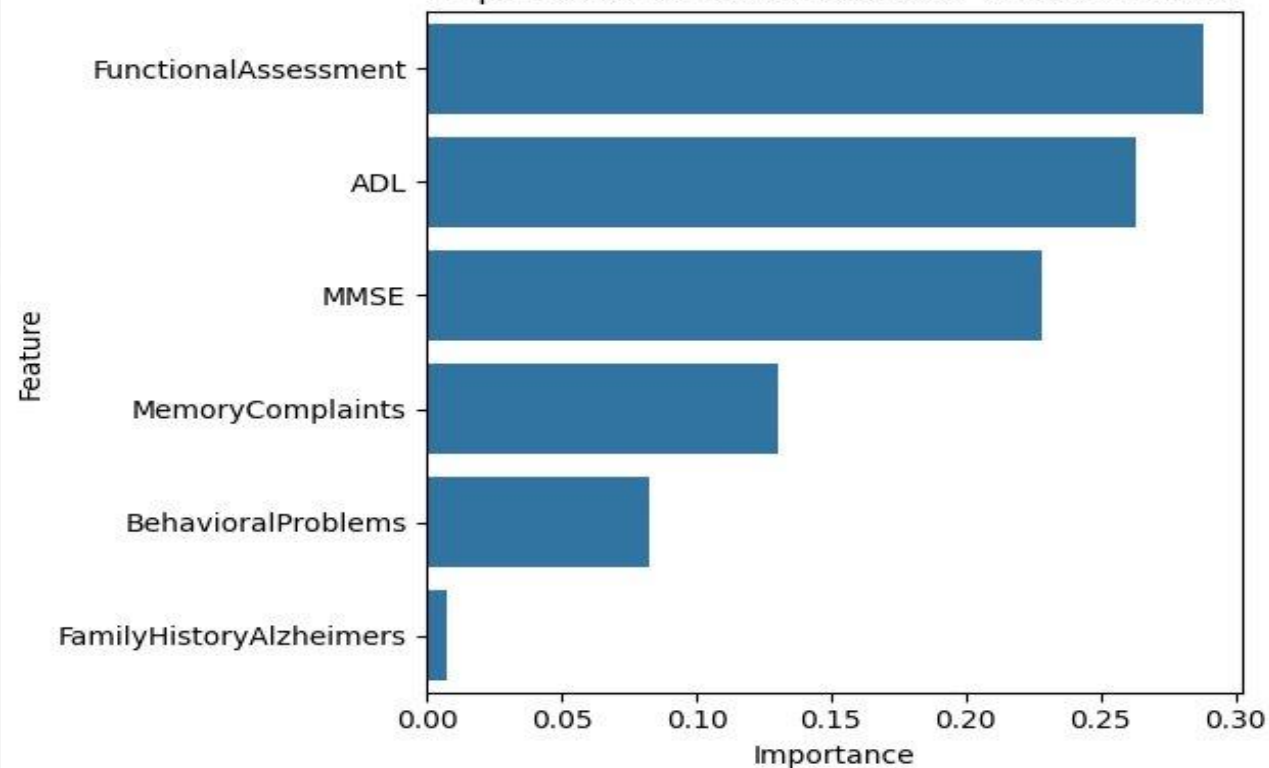


```
# =====  
# 3. Entrenar Random Forest inicial  
# =====  
modelo_rf = RandomForestClassifier(n_estimators=100, random_state=42)  
modelo_rf.fit(X_train, y_train)  
  
# Importancias  
importances = modelo_rf.feature_importances_  
feature_names = X.columns  
feature_df = pd.DataFrame({"Feature": feature_names, "Importance": importances})  
feature_df = feature_df.sort_values(by="Importance", ascending=False)  
  
print("Importancia de variables:\n", feature_df)  
  
# Plot  
sns.barplot(x="Importance", y="Feature", data=feature_df)  
plt.title("Importancia de las variables en Random Forest")  
plt.tight_layout()  
plt.show()
```

Importancia de variables:

	Feature	Importance
1	FunctionalAssessment	0.288187
4	ADL	0.263342
0	MMSE	0.228604
2	MemoryComplaints	0.130639
3	BehavioralProblems	0.082148
5	FamilyHistoryAlzheimers	0.007080

Importancia de las variables en Random Forest

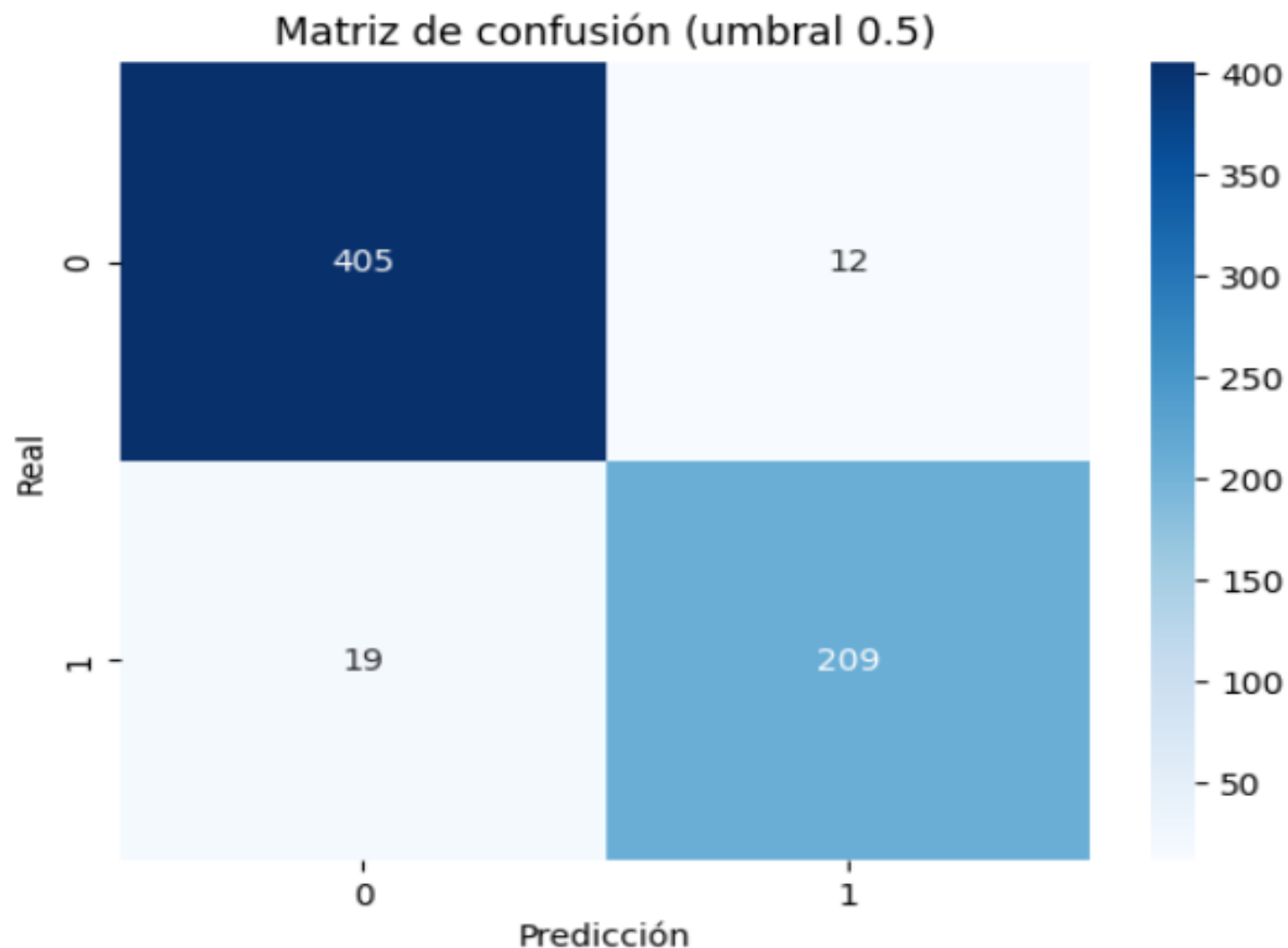




```
# =====  
# 6. Evaluación en test  
# =====  
y_probs = best_rf.predict_proba(X_test)[: , 1]  
y_pred_default = (y_probs >= 0.5).astype(int)  
  
print("\n--- Rendimiento con umbral 0.5 ---")  
print("Accuracy:", accuracy_score(y_test, y_pred_default))  
print("AUC:", roc_auc_score(y_test, y_probs))  
print(classification_report(y_test, y_pred_default))  
  
sns.heatmap(confusion_matrix(y_test, y_pred_default),  
            annot=True, fmt="d", cmap="Blues")  
plt.title("Matriz de confusión (umbral 0.5)")  
plt.xlabel("Predicción"); plt.ylabel("Real")  
plt.show()
```

```
--- Rendimiento con umbral 0.5 ---  
Accuracy: 0.951937984496124  
AUC: 0.946569060541041
```

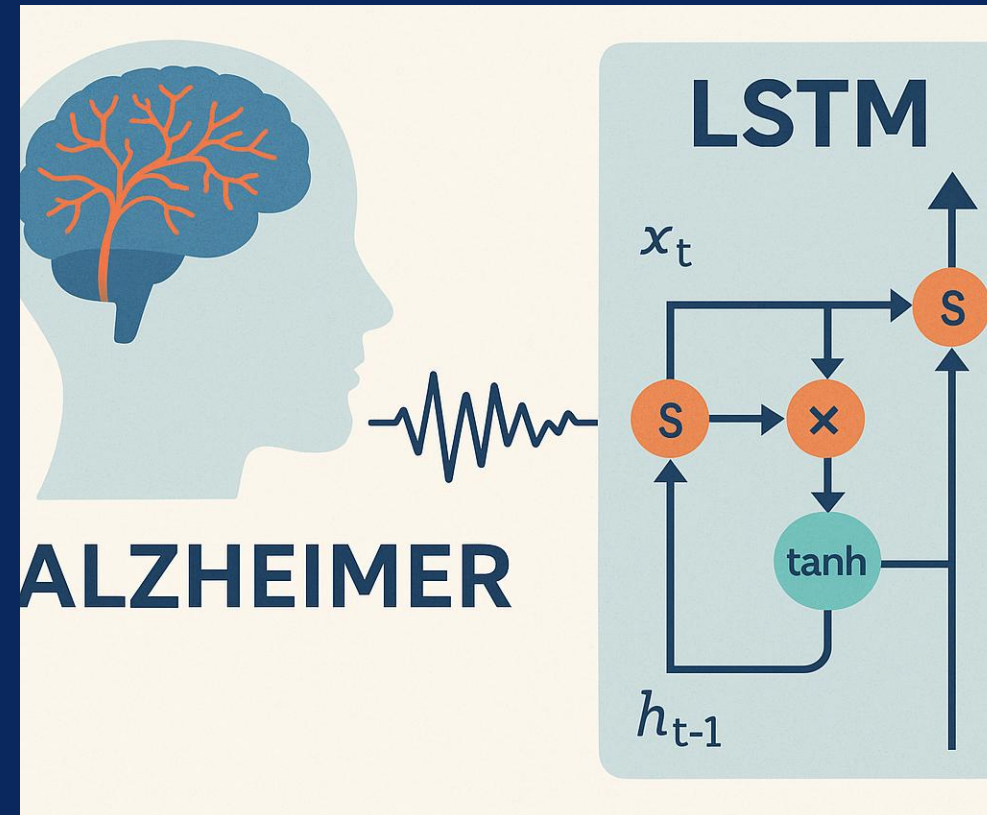
	precision	recall	f1-score	support
False	0.96	0.97	0.96	417
True	0.95	0.92	0.93	228
accuracy			0.95	645
macro avg	0.95	0.94	0.95	645
weighted avg	0.95	0.95	0.95	645





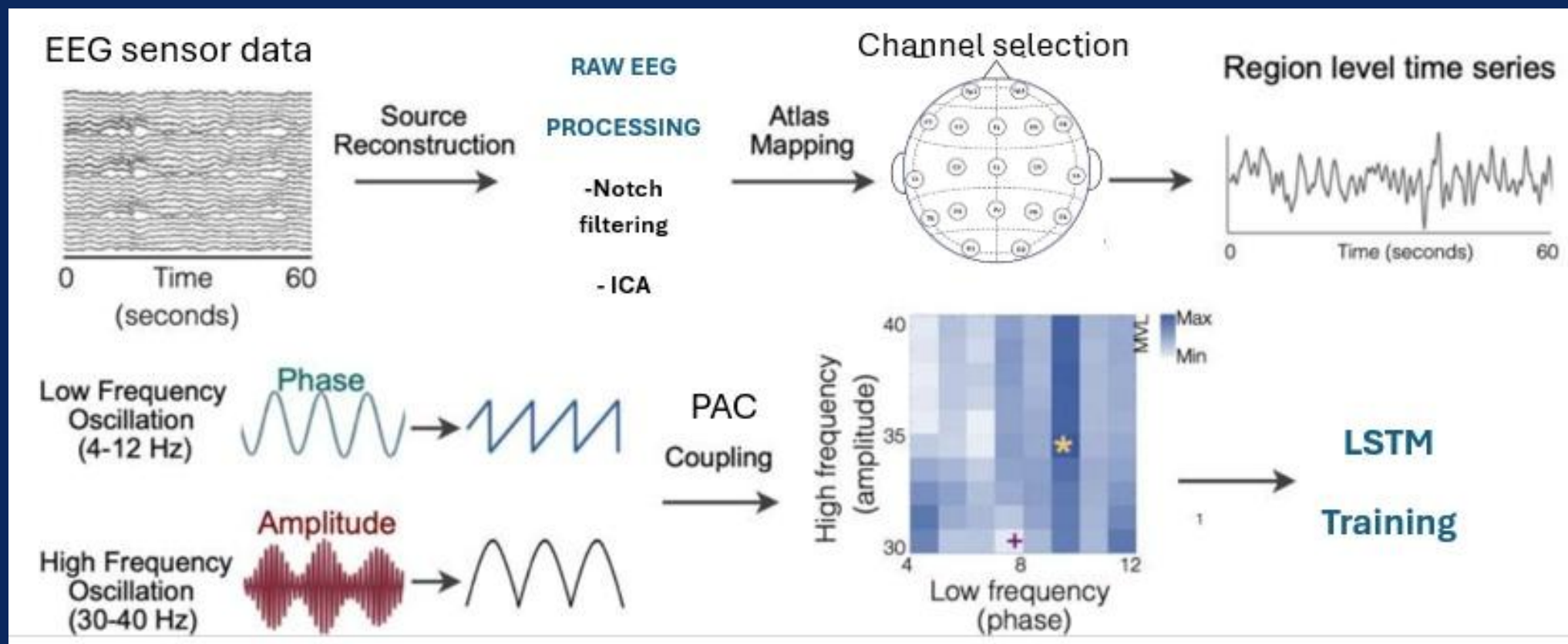
4. METODOLOGÍA

- **PREPROCESAMIENTO** 
Variables: MI_gs, MI_gf, PACz_gs, PACz_gf
Normalización (StandardScaler)
- **MODELO LSTM** 
Capta patrones temporales en las señales
- **ENTRENAMIENTO Y VALIDACIÓN** 
División train/test
Accuracy





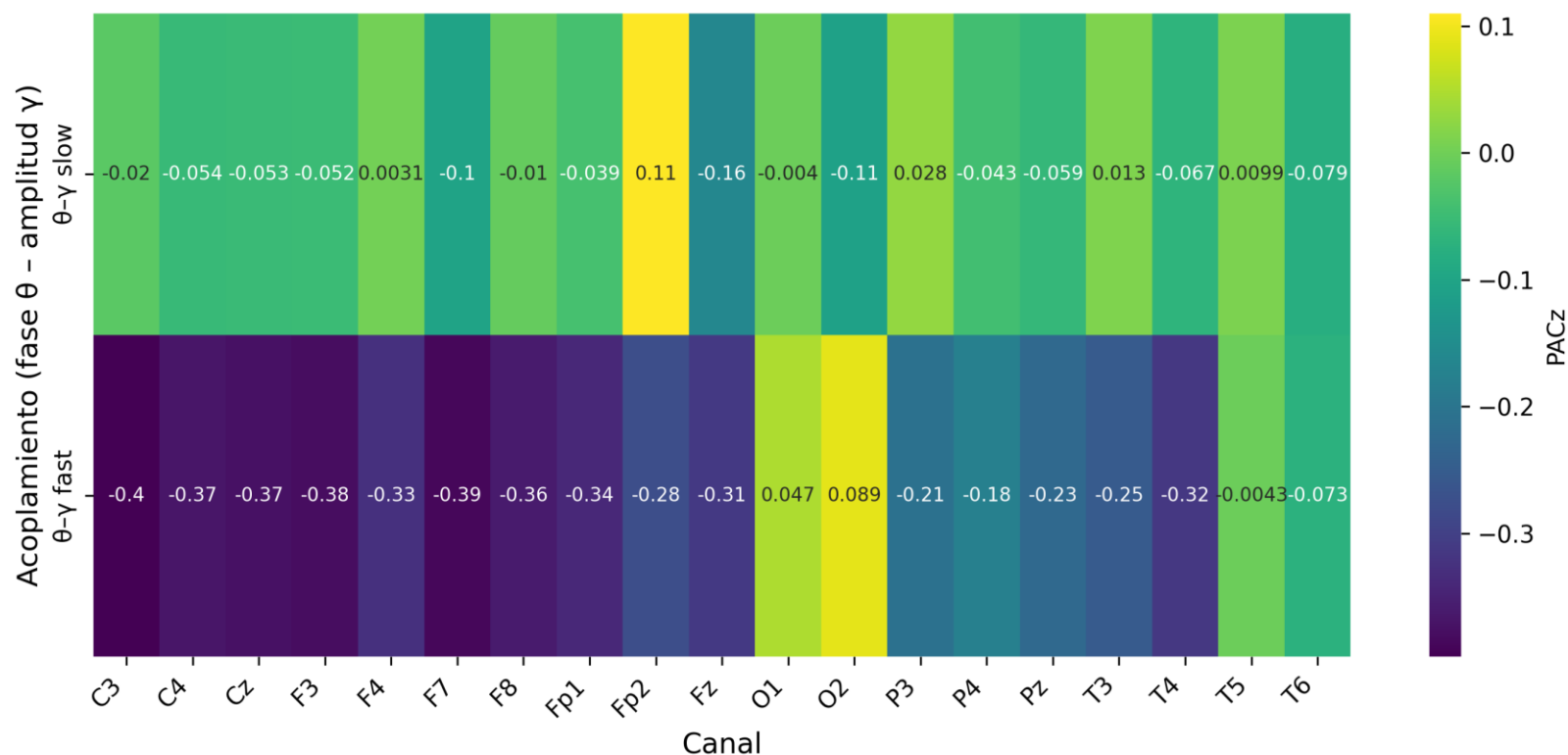
4. METODOLOGÍA





5. RESULTADOS

Comodulograma θ - γ (PACz) - sub-062_task-photomark_eeg



PACIENTE 062
(CONTROL)

VALORES POSTIVOS EN
ACOPLAMIENTO THETA-
GAMMA FAST
(ACOPLAMIENTO
NECESARIO EN
PROCESOS MEMORIA
EN HIPOCAMPO Y
CORTEZA PREFRONTAL)

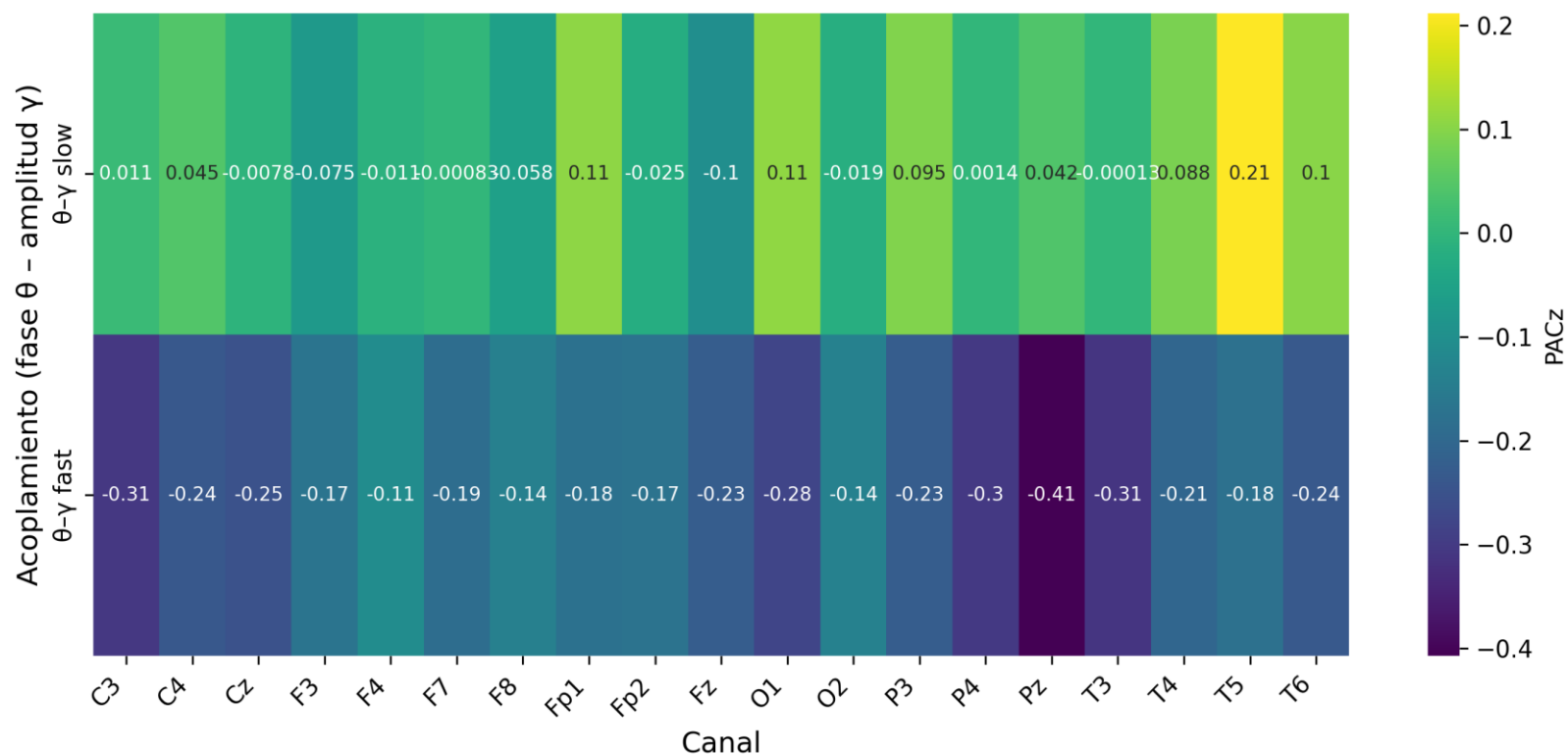
SUJETO OJOS
ABIERTOS





5. RESULTADOS

Comodulograma θ - γ (PACz) - sub-063_task-photomark_eeg



PACIENTE 063
(CONTROL)

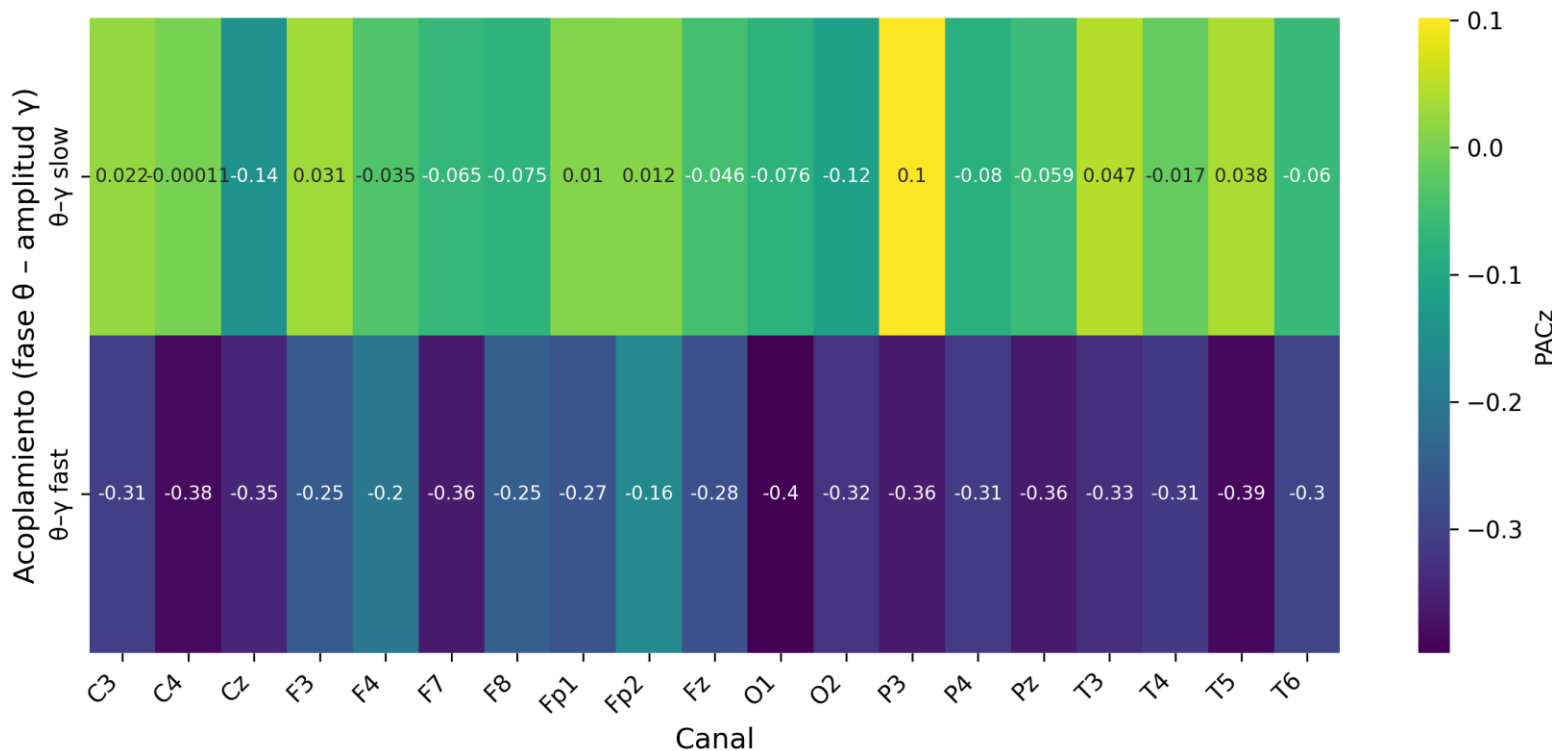
VALORES POCO
NEGATIVOS EN
ACOPLAMIENTO THETA-
GAMMA FAST
(ACOPLAMIENTO
NECESARIO EN
PROCESOS MEMORIA
EN HIPOCAMPO Y
CORTEZA PREFRONTAL)





5. RESULTADOS

Comodulograma θ - γ (PACz) - sub-006_task-photomark_eeg



PACIENTE 006
(ALZHEIMER)
VALORES MUY BAJOS EN
ACOMPLAMIENTO
THETA-GAMMA SLOW
(CERCA 0)
VALORES NEGATIVOS EN
ACOMPLAMIENTO THETA-
GAMMA FAST
(DESACOMPLAMIENTO ,
DISFUNCIÓN
COGNITIVA)





5. RESULTADOS

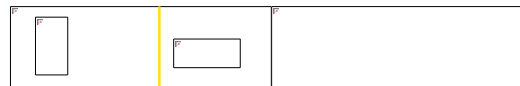
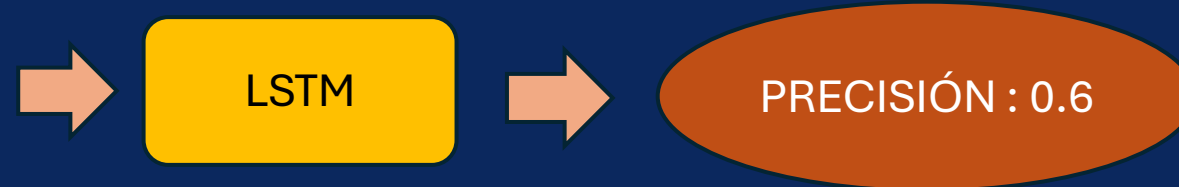
Entrenamiento red neuronal (LSTM)

```

Paciente Canal  MI_gs  MI_gf  PACz_gs \
0  sub-001_task-photomark_eeg (2)  C3  0.002486  0.001236  0.009286
1  sub-001_task-photomark_eeg (2)  C4  0.001301  0.000413  -0.071415
2  sub-001_task-photomark_eeg (2)  Cz  0.002408  0.001183  -0.108514
3  sub-001_task-photomark_eeg (2)  F3  0.002846  0.001317  -0.059137
4  sub-001_task-photomark_eeg (2)  F4  0.002991  0.001254  0.031357
..  ..  ..  ..  ..  ..
869  sub-065_task-photomark_eeg (1)  Pz  0.012987  0.001041  0.016037
870  sub-065_task-photomark_eeg (1)  T3  0.013281  0.001070  0.027425
871  sub-065_task-photomark_eeg (1)  T4  0.013061  0.001066  0.003971
872  sub-065_task-photomark_eeg (1)  T5  0.012352  0.000942  0.045579
873  sub-065_task-photomark_eeg (1)  T6  0.012845  0.000919  0.067688

PACz_gf  Diagnosis  Condición
0  -0.406491  1  ojos_abiertos
1  -0.470645  1  ojos_abiertos
2  -0.459382  1  ojos_abiertos
3  -0.449229  1  ojos_abiertos
4  -0.395775  1  ojos_abiertos
..  ..  ..
869  -0.363551  0  ojos_abiertos
870  -0.343203  0  ojos_abiertos
871  -0.366529  0  ojos_abiertos
872  -0.354248  0  ojos_abiertos
873  -0.273482  0  ojos_abiertos

[874 rows x 8 columns]
```





5. RESULTADOS

Mejora del modelo LSTM

- Ojos abiertos
- Ojos cerrados
- Ojos abiertos + ojos cerrados



PRECISIÓN :
0.54-0.6

NO NORMALIZAR

```
import torch.nn as nn

class LSTMClassifier(nn.Module):
    def __init__(self, input_dim=4, hidden_dim=64, num_layers=1, num_classes=2):
        super(LSTMClassifier, self).__init__()
        self.lstm = nn.LSTM(input_dim, hidden_dim, num_layers, batch_first=True)
        self.fc = nn.Linear(hidden_dim, num_classes)

    def forward(self, x):
        # x: (batch, seq_len, input_dim)
        _, (hn, _) = self.lstm(x) # hn: (num_layers, batch, hidden_dim)
        out = self.fc(hn[-1]) # tomamos la última capa
        return out
```



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5. RESULTADOS

Perspectivas futuras

1. Ajustes preprocesamiento
2. Empleo de EGG combinado con test de memoria (MSIT, MMSE, Sternberg)

A Polish Electroencephalography, Alzheimer's Risk-genes, Lifestyle and Neuroimaging (PEARL-Neuro) Database

IMPORTANT NOTE: The dataset contains no errors (BIDS-1). The numerous warnings currently displayed are a result of OpenNeuro updating its validator to BIDS-2. The OpenNeuro team is actively working on refining the validator to display only meaningful warnings (more information on OpenNeuro GitHub page). At this time, as dataset owners, we are unable to take any action to resolve these warnings.

Data Descriptor:

- doi.org/10.1038/s41597-024-03106-5 (<https://www.nature.com/articles/s41597-024-03106-5>)

